

Saber Ayatollahi, CFA, MSc

Toronto, Canada • +1 (416) 856-1276 • saber.ayatollahi@gmail.com • www.linkedin.com/in/sayatollahi

SUMMARY

Balance-sheet and liquidity risk professional with ~7 years across ALM, IRRBB analytics, and model validation and governance. At Moody's Analytics, holds delegated sign-off authority on valuation, sensitivity, and ALM outputs for institutional portfolios (\$5-25bn per engagement) – independently reviewing model methodology, curve construction, spread calibration, and assumption defensibility prior to release, and benchmarking outputs against economic intuition and alternative calibrations. Prior insurance-accounting transformation at EY and stochastic ALM/LDI studies for pension funds at Ortec Finance. CFA charter, dual MSc (Financial Modelling; Chemical Engineering).

CORE SKILLS

ALM & IRRBB Analytics • Model Validation & Independent Review • Liquidity Gap & Cash Flow Projection • Curve Construction & Spread Calibration • Sensitivity & Scenario Analysis • Stochastic Modelling & Monte Carlo • Model Governance & Documentation • Python, SQL, Excel • CFA Charterholder

PROFESSIONAL EXPERIENCE

Moody's Analytics

Asst. Director, Modelling Services | Toronto, May 2023 - Present

Model Validation & Independent Review

- Hold delegated sign-off authority on valuation, sensitivity, and ALM outputs delivered to institutional client portfolios ranging \$5-25bn per engagement (cumulatively \$50bn+) – independently review model methodology, curve construction, spread calibration, assumptions, and cross-asset interactions prior to production release.
- Operate within a formal model-governance framework: challenge methodology choices, validate economic defensibility of assumptions, benchmark model performance against alternative calibrations and economic intuition, and document findings to standards consistent with SR 11-7 and OSFI E-23 expectations.
- Escalate outputs lacking mathematical or economic defensibility; serve as escalation point for client-critical analytical issues, interfacing with clients' Heads of Risk and investment stakeholders.
- Review aggregation logic converting security-level exposures into portfolio-level risk metrics feeding downstream ALM processes.

Balance Sheet, IRRBB & Liquidity Analytics

- Oversee interest rate risk and duration analysis under parallel and non-parallel rate shocks – including sensitivity analysis at portfolio-level ALM aggregates – aligned with IRRBB standards analogous to OSFI B-12 and Basel Committee frameworks.
- Led design and implementation of an enterprise multi-asset cash flow projection engine supporting base, stress, and reverse-stress scenarios – delivering forward-looking liquidity visibility across ALM horizons.
- Architected configurable time-bucketed liquidity gap analytics (T+1 through multi-year) enabling funding optimization, refinancing risk assessment, and asset-allocation analysis.
- Embedded behavioral cash flow assumptions, prepayment logic, and macro stress overlays aligned with regulatory liquidity expectations.
- Validated derivatives pricing outputs (rates, FX, inflation) and cross-checked sensitivity consistency at portfolio-level ALM aggregates.

Analytics Engineering

- Re-engineered manual spreadsheet workflows into scalable, auditable Python analytics pipelines with embedded logging and validation controls – supporting model-governance audit readiness.
- Prepared analytical summaries for senior stakeholders covering interest-rate exposure, scenario impacts, assumption sensitivity, and balance-sheet implications.

Client Service Specialist | Toronto, May 2022 - May 2023

- Led onboarding and implementation for U.S. and Canadian pension funds, asset managers, and consulting firms migrating to the PFaroe analytics platform – advised on asset, liability, and ALM configurations and validated model outputs post-deployment.
- Served as technical liaison between client investment teams and product owners; scoped requirements into development requests and contributed to successful migration of all assigned accounts from the legacy Calypso platform.

Ernst & Young (EY)

Senior Consultant, IFRS & Financial Risk Transformation | Toronto, Sep 2021 - Apr 2022

- Delivered IFRS 17 and IFRS 9 technology-transformation projects for Canadian and international insurance clients – aligned regulatory requirements with system implementation and financial-reporting frameworks.
- Advised actuarial, IT, finance, and program-management functions on adoption of IFRS 17/9 standards; contributed to governance documentation and regulatory readiness processes.

Ortec Finance Canada

Consultant, Asset-Liability Management & Risk | Toronto, Feb 2019 - Sep 2021

- Delivered ALM studies for Canadian and international pension funds and institutional investors – analyzed funding ratios, duration mismatch, and long-term balance-sheet risk exposures.
- Built and interpreted stochastic economic scenario generators assessing interest rate risk and funding volatility under base-case and stressed environments.
- Conducted liability-driven investment (LDI) analysis and cash flow, liquidity, and hedging analysis across interest-rate, inflation, and currency risk for institutional clients.
- Performed asset-only and asset-liability (surplus) portfolio optimization on VaR and CVaR using Ortec's GLASS platform; conducted risk decomposition and contribution-to-risk attribution.
- Led onboarding and the ALM study for a three-plan university pension merger (UPP) – developed and validated the model capturing Funding Policy dynamics, duration, currency, inflation, and leverage overlay strategies.

EDUCATION & CREDENTIALS

- Chartered Financial Analyst (CFA) - CFA Institute, 2024
- MSc, Financial Modelling - Western University, 2018
- MSc, Chemical Engineering (Full Scholarship) - Western University, 2016